

Confidence limit for Proportions

By Central Limit Theorem, we have

$$Z = \frac{t - E(t)}{SE(t)} \sim N(0,1)$$

Substitute $t=p$ where p is the proportion.

$$\text{then } Z = \frac{p - E(p)}{SE(p)} \sim N(0,1)$$

We have $E(p) = P$ & $SE(p) = \sqrt{PQ/n}$

where $Q = 1 - P$

$$\text{ie } Z = \frac{p - P}{\sqrt{PQ/n}} \sim N(0,1)$$

$100(1-\alpha)\%$ is ^{the} given ~~by~~ confidence limit for p

Confidence Interval of proportion is

$$p \pm Z_{\alpha/2} \sqrt{PQ/n}$$

$$95\% \text{ CI is } p \pm 1.96 \sqrt{PQ/n}$$

$$99\% \text{ CI is } p \pm 2.58 \sqrt{PQ/n}$$

Notes :- Confidence interval for proportion

→ If no estimate of population proportion 'p' is available, in such situations $p=0.5$ & $q=0.5$.

$$SE = \sqrt{\frac{pq}{n}}$$

$$= \sqrt{\frac{0.5 \times 0.5}{n}}$$

$$= \frac{0.5}{\sqrt{n}}$$

→ In case of confidence limit for proportion, if the population size N is given, then the standard error is given by,

$$SE(p) = \sqrt{\frac{(N-n)pq}{N(n-1)}}$$

Or in special cases,

$$SE(p) = \sqrt{\frac{(N-n)pq}{n(N-1)}}$$

where N is the population size,

n is the sample size,

p is the proportion

$$q = 1 - p.$$

The confidence limits can be found out by,

$$p \pm Z_{\alpha/2} \sqrt{\frac{(N-n)pq}{N(n-1)}}$$

and this is applicable for simple random sampling without replacement.

Confidence limit for difference of proportions

If p_1 and p_2 are the sample proportions in a large and independent random samples of sizes n_1 and n_2 respectively from 'infinite' population with proportions P_1 and P_2 , then

$$E(p_1 - p_2) = E(p_1) - E(p_2) = P_1 - P_2$$

$$\text{Var}(p_1 - p_2) = \text{Var}(p_1) + \text{Var}(p_2)$$

$$= \frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}$$

$$SE(p_1 - p_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

Confidence limits for difference of proportion is given by,

$$(p_1 - p_2) \pm Z_{\alpha/2} \cdot SE(p_1 - p_2)$$

$$\text{i.e., } (p_1 - p_2) \pm Z_{\alpha/2} \cdot \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$